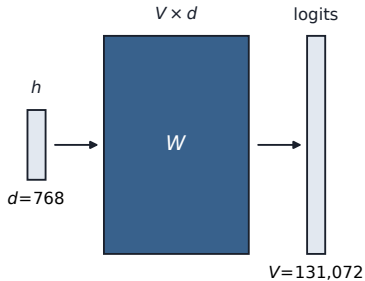


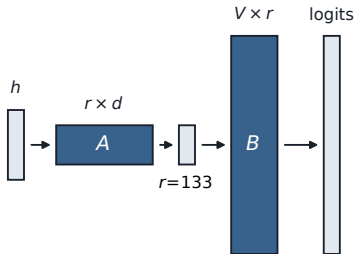
(a) dense head



every word reads all d features

FLOPs/token	100.7M
matrix rank	768
views x capacity	1 x 768

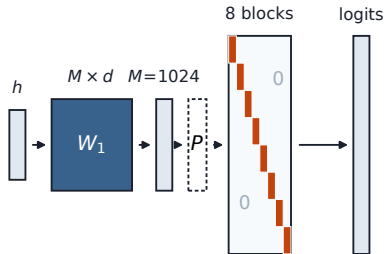
(b) low-rank head, cost-matched



every word reads the same r features

FLOPs/token	17.5M
matrix rank	133
views x capacity	1 x 133

(c) Monarch head (ours)



each word reads only its block's $m_1=128$ features

FLOPs/token	17.6M
matrix rank	768
views x capacity	8 x 128